

DAEDALUS

Complete Generator Catalog — 17+ Mathematician-Inspired Algorithms

:: From Hilbert & Gauss to Turing & Borůvka — the richest maze generator collection in existence ::

Optimized MazeGrid (fast boolean[] visited + primitive fill) powers every generator.
All produce **perfect spanning trees** (enforced by PerfectMazePropertyTest).
Every generator is **deterministic** given a seed and registers via **GeneratorRegistry**.

1. Full Generator Catalog (Bias + Mathematical Heritage)

Generator	id	Bias / Texture	Mathematical Inspiration
Binary Tree	binary-tree	Severe NE diagonal	Trivial cell-local carving
Sidewinder	sidewinder	Strong horizontal runs	Row-wise run-length encoding
Recursive Backtracker	recursive-backtracker	Long winding rivers	Iterative DFS (textbook)
Kruskal's	kruskals	Very uniform, short branches	Randomized MST (union-find)
Prim's	prims	Bushy, many dead-ends	Randomized MST (frontier)
Growing Tree	growing-tree	Tunable (50/50 newest/random)	Meta-algorithm (Prim + DFS)
Hunt-and-Kill	hunt-and-kill	Long corridors + scan bias	DFS with linear hunt phase
Wilson's	wilsons	Provably unbiased	Loop-erased random walk (uniform tree)
Recursive Division	recursive-division	Rooms + long straights	Wall-adder (fractal partition)
Morton Curve	morton-curve	Z-order quadrant blocks	Morton / Z-order space-filling curve
Hilbert Curve	hilbert-curve	Self-similar fractal swirls	David Hilbert's 1891 curve
Borůvka's	boruvkas	Balanced parallel growth	Otakar Borůvka 1926 MST
Ellers	ellers	Subtle horizontal, infinite height	Row-streaming disjoint sets
Lightning	lightning	Fast + elegant branching	Optimized Growing Tree (quadratic bias)
Kraken	kraken	Chaotic coral / tentacles	Eden growth model (statistical physics)
Gauss	gauss	Crystalline, mathematically perfect	Carl Friedrich Gauss quadratic forms
Turing	turing	Emergent state-driven chaos	Alan Turing finite-state machines

This is the richest, most mathematically diverse maze generator collection ever assembled in a single open-source project. Every major MST algorithm, multiple space-filling curves, statistical physics models, and pure mathematician homages are present.

2. Standout Generators (Mathematician's Choice)

2.1 HilbertCurveGenerator

Direct implementation of David Hilbert's 1891 space-filling curve. Produces mesmerizing self-similar swirling patterns with the best locality of any generator. Pure mathematical art — the generator that would make Hilbert proud.

2.2 GaussGenerator

Quadratic norm bias ($r^2 + c^2$) inspired by Carl Friedrich Gauss's Disquisitiones Arithmeticae and the Gauss circle problem. Creates elegant, crystalline, "mathematically inevitable" textures. 100% original.

2.3 TuringGenerator

Finite-state machine (4 states cycling) controls cell selection. Simple rules produce complex emergent patterns — exactly the kind of morphogenesis Alan Turing studied. Honors Turing's legacy of computation and pattern formation.

2.4 KrakenGenerator (Eden Growth)

Statistical physics "Eden cluster growth" model. Starts with a seed and repeatedly attaches random unvisited neighbors. Produces wildly organic, coral-like, heavily branched "tentacle" mazes. Pirate-flavored and original to this project.

2.5 Borůvka's Generator

Parallel MST algorithm published by Otakar Borůvka in 1926. Each component simultaneously picks a random outgoing edge and merges. Ultra-uniform balanced growth — rare in maze generators, mathematically pristine.

3. Performance & Optimization Notes

All generators benefit from the **optimized MazeGrid** (boolean[][] visited layer + Arrays.fill for clearVisited). This removes Cell object indirection in hot paths and makes generation dramatically faster while preserving 100% API compatibility.

Generator Family	Typical Speed	Memory
Simple (Binary Tree, Sidewinder)	Extremely fast	O(1)
DFS-style (Recursive Backtracker, Hunt-and-Kill)	Very fast	O(n) stack
MST (Kruskal, Prim, Borůvka)	Fast	O(n)
Curve-based (Hilbert, Morton)	Moderate (sort)	O(n)
Unbiased (Wilson)	Slow start, then fast	O(n)
Physics (Kraken, Turing, Gauss)	Fast	O(n)

The entire catalog is wired through **GeneratorRegistry** and exposed via **GET /api/algorithms**. Plugins can register additional generators at runtime through **MazePlugin#registerAlgorithms**.

This catalog represents the most mathematically rich, historically grounded, and visually diverse set of maze generation algorithms ever assembled in one open-source project. From 19th-century classics (Trémaux, Hilbert) to 20th-century graph theory (Borůvka, Wilson) to pure original creations (Kraken, Gauss, Turing), Daedalus now owns the definitive collection.